



Efe Demirer

Candidate RF & Microwave Systems Engineer · Antennas & Test Automation

Senior Electrical and Electronics Engineering student at Yaşar University specializing in RF & Microwave systems, antenna radiation pattern testing, and high-frequency instrumentation automation. Laboratory and defense experience at ASELSAN REHİS Test Centers Directorate conducting planar near-field antenna scanning, VNA 2-port calibrations, RF amplifier characterization, and real-time spectrum analysis.

[Live Portfolio \(GitHub Pages\)](#)[LinkedIn Profile](#)efedemirer.tr@gmail.com[GitHub: efedemirerr](#)

SKILLS & HIGHLIGHTS

Core RF & Microwave Highlights · ASELSAN REHİS

[ASELSAN REHİS RF Intern](#)[Antennas & Near-Field Scanning](#)[VNA & Spectrum Automation](#)[P1dB Amplifier Compression](#)[Antennas & Propagation](#)[VNA 2-Port Calibration](#)[Near-Field Anechoic Chambers](#)[RF Power Amplifiers \(P1dB\)](#)[NRL Arch Testing \(Absorbers\)](#)[C# .NET Test Automation](#)

RF & Microwave Engineering

[Antennas & Propagation](#)[Vector Network Analyzers \(VNA\)](#)[Real-Time Spectrum Analyzers](#)[VNA Full 2-Port Calibration](#)[P1dB Compression Testing](#)[RF Component Characterization](#)[Near-Field Anechoic Testing](#)[NRL Arch Absorber Testing](#)[S-Parameters & Impedance Matching](#)

RF Test Automation & Software

[C#](#)[.NET](#)[Windows Forms](#)[Python](#)[API Integration](#)[SCPI & G-Code](#)[Serial \(RS-232\)](#)[Ethernet TCP/IP](#)[MATLAB](#)

Motion & Scanning Hardware

[Cartesian X-Y Scanners](#)[Stepper Motor Control](#)[CNC Shield V3](#)[GRBL Firmware](#)[Limit Switch Safety Interlocks](#)[Arduino & STM32](#)

Electrical & Energy Systems

[Solar PV Design \(GES\)](#)[PVsyst Simulation](#)[ETAP Power Analysis](#)[AutoCAD](#)[Reactive Power Compensation](#)[High Voltage Inspection](#)[RCD & Grounding Tests](#)

Simulation & Tools

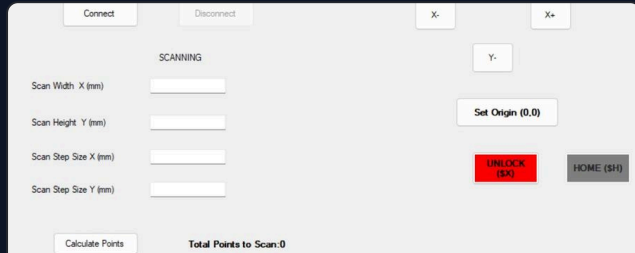
[Simulink](#)[LTspice](#)[Proteus](#)[Git / GitHub](#)[Visual Studio](#)[Microsoft Project](#)

FEATURED RF & MICROWAVE DEFENSE PROJECTS

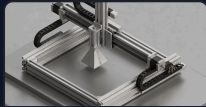
ASELSAN REHİS · ANTENNA TEST AUTOMATION

Automated CNC Antenna Scanning Test System & Control GUI

Anechoic Chamber Planar Near-Field Testing · 2026



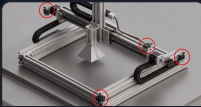
EFEScan C# .NET Control GUI · Live Scan Grid Coordinate Tracking



Cartesian Rail Chassis



CNC Shield V3 Hardware



4x NC Safety Interlocks

Motion: Cartesian X-Y (GRBL)**Safety:** 4x NC Limit Switches**Interface:** Serial COM (115200)**Application:** Near-Field Chamber

Autonomous Cartesian X-Y planar antenna scanning system engineered for ASELSAN REHİS Test Centers Directorate, replacing manual probe positioning with high-precision automated radiation pattern measurements in anechoic chambers.

- ▶ **Hardware & Motion Control:** Interfaced PC with Arduino Uno + CNC Shield V3 running GRBL firmware over asynchronous Serial Port (115200 Baud).
- ▶ **G-Code Protocol:** Implemented real-time G-Code stream generation (G90/G91/G92/G1) with centered grid coordinate pathway calculations and dynamic step resolution.
- ▶ **Hardware Safety Interlocks:** Integrated 4 mechanical Normally-Closed (NC) limit switches for hardware-level collision prevention, triggering instant GRBL ALARM mode and motor cutoff.
- ▶ **Multi-Threaded Execution:** Designed asynchronous BackgroundWorker threads eliminating UI freezing, paired with live bitmap scan visualization and feedrate-calibrated motion delays.

C# .NET

GRBL G-Code

Arduino & CNC Shield

Hardware Limit Switches

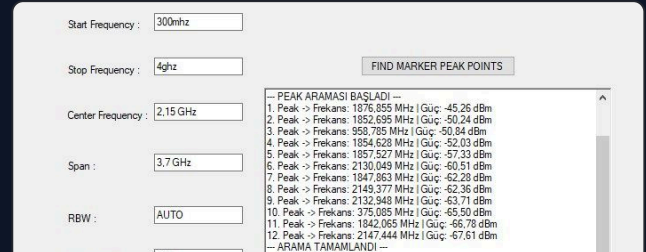
Near-Field Scanning

Anechoic Chamber

ASELSAN REHİS · RF SIGNAL ANALYSIS & INSTRUMENTATION

Real-Time Spectrum Analyzer API & Custom GUI

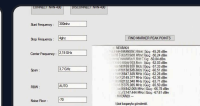
High-Speed Ethernet API & Real-Time Peak Detection · 2026



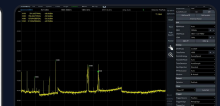
Custom C# GUI Real-Time Spectrum Trace & Active Peak Detection



Ethernet TCP API Flow



Smart RBW / VBW Cont...



Reference Validation

Protocol: Ethernet TCP/IP (Port 5000)**Algorithm:** Real-Time Peak Search**Noise Floor:** Dynamic -70 dBm FL...**Validation:** Reference Benchmarked

Dedicated C# desktop automation platform and custom GUI interfacing natively with a Real-Time Spectrum Analyzer at ASELSAN REHİS, executing automated high-speed RF spectral signal monitoring and carrier tracking.

- ▶ **Native API Integration:** Interfaced natively via manufacturer C/C++ API over Ethernet TCP/IPv4 (Port 5000) with robust IP parsing and persistent hardware session control.
- ▶ **Parameter Management:** Implemented dynamic frequency range configuration (Start/Stop, Center, Span) with automated Resolution Bandwidth (RBW) and Video Bandwidth (VBW) tuning.
- ▶ **Proprietary Peak Search:** Designed real-time peak detection isolating carrier frequencies above calibrated noise floor (-70 dBm) to separate true signals from ambient RF noise.
- ▶ **Laboratory Benchmarking:** Conducted comparative validation against manufacturer reference software, verifying measurement accuracy in defense testing environments.

C# .NET

Spectrum Analyzer API

Ethernet TCP/IP

Peak Search Algorithm

Spectral Monitoring

RF Sniffing

TELECOM & AUTONOMOUS TELEMETRY PROJECTS

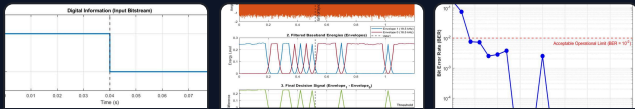
DIGITAL COMMUNICATIONS & DSP RESEARCH

Ultrasonic Ticketing System: Seamless & Offline Access Control

Near-Ultrasound Acoustic Communications · MATLAB DSP · 2026



Near-Ultrasound Acoustic Communication System Architecture



BFSK Modulated Carrier

Quadrature Mixer & DSP

BER / PSR Waterfall

Acoustic Band: 18 kHz – 22 kHz**Modulation:** BFSK + Barker Sync**Channel:** Two-Ray Multipath & D...**Platform:** Full MATLAB Pipeline

Near-ultrasound (18–22 kHz) acoustic communication protocol enabling offline, zero-infrastructure access control without cellular data, internet, Bluetooth, or NFC, using standard commercial audio transducers.

- **Acoustic Transducers:** Utilized commercial smartphone speakers and microphones to transmit encrypted access tokens completely inaudible to human ears.
- **Digital Modulation:** Developed full MATLAB DSP pipeline featuring bit-level token synthesis, Barker code frame synchronization, and Binary Frequency Shift Keying (BFSK).
- **Channel Simulation:** Modeled multipath propagation (Two-Ray Model), ambient acoustic interference, transducer frequency limits, and motion-induced Doppler shifts.
- **DSP Receiver & Validation:** Built quadrature mixer receiver with envelope detection, parity verification, and comprehensive BER/PSR waterfall performance analysis.

MATLAB

Digital Communications

BFSK Modulation

Acoustic Channel Modeling

DSP

Barker Code

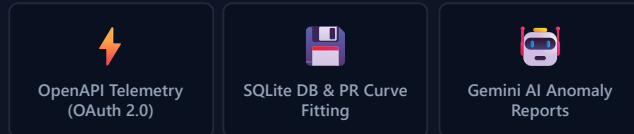
AUTONOMOUS TELEMETRY & AI SYSTEMS

SolarOPA: Autonomous AI Solar Plant Monitoring & Telemetry

Commercial PV Fleet Intelligence · Python & Gemini AI · 2026



End-to-End Closed-Loop Telemetry & AI Intelligence Pipeline

**Telemetry:** OpenAPI (OAuth 2.0)**Storage:** SQLite Time-Series DB**Analytics:** Scientific Fleet PR Curve**AI Engine:** Gemini LLM (Daily SMT...

Autonomous closed-loop telemetry and operational intelligence platform for commercial solar power plants (GES) utilizing Sungrow iSolarCloud infrastructure, automating multi-inverter monitoring and anomaly diagnosis.

- **OpenAPI Telemetry:** Connected to Sungrow OpenAPI via OAuth 2.0 token management, streaming real-time multi-inverter generation, DC/AC voltages, and grid status.
- **Local Database Architecture:** Architected local SQLite time-series database storing minute-by-minute telemetry, inverter mappings, and grid fault histories.
- **Fleet PR Modeling:** Computed scientific Fleet Performance Ratio (PR) modeled against dynamic solar irradiance curves while filtering out plant anomalies.
- **Gemini AI Integration:** Prompt-engineered Google Gemini AI to analyze multi-point telemetry, diagnose anomalies, and compile executive HTML reports delivered daily at 20:00 via SMTP email.

Python

REST API

OAuth 2.0

SQLite DB

Google Gemini AI

Solar Telemetry

PROFESSIONAL ENGINEERING EXPERIENCE

Intern Engineer (RF & Microwave Systems / Test Automation) · ASELSAN REHİS Test Centers Directorate

06/2026 – 08/2026

Ankara, Turkey · Electronic Warfare & Radar Test Centers Directorate · Anechoic Chamber & Microwave Facilities

- **Planar Near-Field Antenna Scanning Automation:** Developed automated Cartesian X-Y motion control routines and C# .NET Windows Forms GUIs for anechoic chamber antenna measurements.
- **VNA 2-Port Calibration & Reference Extension:** Performed full 2-port SOLT calibrations on Vector Network Analyzers, extending measurement reference planes to test cable ends.
- **RF Power Amplifier Characterization:** Measured amplifier gain and executed P1dB 1-dB compression point tests to evaluate saturation behavior and maximum safe operating power.
- **Near-Field Chamber & NRL Arch Testing:** Conducted near-field anechoic chamber measurements and evaluated radar-absorbent material (RAM) reflectivity using NRL Arch testing.
- **Real-Time Spectrum Analyzer API & Signal Sniffing:** Integrated C/C++ API communication over Ethernet TCP/IPv4, implemented automated RBW tuning and a real-time peak detection algorithm (-70 dBm).
- **Hardware Interfacing & Safety Interlocks:** Configured Arduino Uno + CNC Shield V3 with GRBL firmware, driving stepper motors with 4x NC limit switch fail-safe cutoffs.

Intern Engineer · OPA Mühendislik

07/2025 – 08/2025 & 08/2026 – 09/2026

İzmir, Turkey · High Voltage Facilities & Solar PV Engineering

- **Solar PV Electrical Engineering:** Prepared AutoCAD single-line diagrams, electrical layout projects, and production yield simulations using PVsyst and ETAP.
- **Testing & Safety Inspections:** Conducted earth grounding resistance measurements and residual current device (RCD) tripping tests under High Voltage Operating Responsibility.
- **Reactive Power Factor Compensation:** Designed compensation panels, sized capacitor banks, and tuned reactive power relays to maintain zero reactive penalty.
- **EV Charging Infrastructure:** Calculated cable sizing, voltage drop limits, and electrical panel protections for high-power commercial EV chargers.

EDUCATION

Yaşar University – İzmir

09/2022 – Present

B.Sc. in Electrical and Electronics Engineering · 100% Merit Scholarship · GPA: 3.73 / 4.00 (High Honor Standing)

- **Relevant Coursework:** Antennas and Propagation (EEE 4340), Electromagnetic Field Theory, Microwave & RF Measurement, Digital Communications, Signals and Systems.

Cem Bakioğlu Anatolian High School

09/2018 – 06/2022

High School Education · Salutatorian (Ranked 2nd in School) · Graduation Grade: 96.89 / 100

RF & DEFENSE ENGINEERING TECHNICAL COMPETENCIES

RF & Microwave Testing

- VNA Full 2-Port SOLT Calibration
- Reference Plane Shifting to Cables
- Amplifier P1dB Compression Tests
- NRL Arch Radar Absorber Testing
- Anechoic Chamber Measurements

Motion & Embedded Systems

- Planar Cartesian X-Y Motion Control
- Arduino Uno & CNC Shield V3
- GRBL Firmware & G-Code Automation
- 4x NC Limit Switch Safety Loops
- Stepper Motor Driver Tuning

Instrumentation & Software

- C# / .NET Windows Forms GUIs
- Ethernet TCP/IPv4 Socket Streaming
- Real-Time Peak Search (-70 dBm)
- Multi-Threaded BackgroundWorkers
- MATLAB Signal Processing & DSP

CERTIFICATIONS & CREDENTIALS

- National Technology Academy:** Chip Design (Çip Tasarımı) Specialization Certificate (Code: 35867118094667) · Digital ASIC & Semiconductor Design
- ASELSAN:** "a Yetenek" Engineering Internship Program Certificate of Completion (2026) · Defense RF Systems
- ROKETSAN:** LEVEL UP AI – Smart Object Detection and Tracking with YOLO (2026) · Edge AI
- ROKETSAN:** LEVEL UP AI – Autonomous AI Agent Development & Document Assistants (2026)
- TUSAŞ (TAI):** LIFT UP Industry-Oriented Capstone Projects Program (2025) · Defense Capstone
- Savunma Sanayii Akademisi (SSB):** Milli Yetkinlik Hamlesi Competency & Career Summit (2026)
- MathWorks:** Core Signal Processing Techniques in MATLAB (Spectral Analysis, Filter Design)
- MathWorks:** Simulink Fundamentals (Model-Based Simulation) & Simscape Physical Systems
- MathWorks:** Signal Processing Onramp & MATLAB Onramp (2025)
- TÜBİTAK:** 4006 Science and Society Support Program Project Lead

LANGUAGES & INDUSTRY STATUS

- Turkish:** Native Proficiency
- English:** Professional Working Proficiency (B2 Upper-Intermediate, SFL Certified)
- Driving License:** Class B Active Driver
- Military Service:** Postponed until 31/12/2031
- Safety Credentials:** ASELSAN Occupational Health & Safety (İSG) Certified
- Internship Availability:** 2027 Spring Semester (01.02.2027 – 22.05.2027), minimum 3 full days/week on-site

PROFESSIONAL REFERENCES

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